Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Activation steering has recently been shown to reproduce a known protein property (thermostability) in ESM2-650M via difference-of-means vectors. We ask two questions this raises but does not answer: does a scoring proxy's correlation with real experimental labels predict whether steering toward it actually works, and can a single steering run be trusted at face value? Under a pre-registered six-criterion protocol, we test five novel target properties (binding affinity, catalytic activity, intrinsic disorder, immunogenicity, expression yield) and find no positive relationship between proxy validity and causal effect: our best-validated proxy (binding affinity, $r = 0.80$ held-out) steers nothing, while a substantially weaker one (catalytic activity, $r = 0.22$) produces a robust, three-seed-replicated effect. This mirrors an independent, component-level finding in the same model family: attention heads ranked by correlation with real 3D contacts show near-zero relationship to their causal ablation effect. We further show that a single-seed steering result can mislead in criterion-specific ways – one target's effect direction replicates across three seeds while its residue-exclusion artifact check does not (2 of 3) – and introduce a lightweight diagnostic, steering-vector cosine similarity across targets, that explains an otherwise-unexplained ambiguous result as a geometric echo of a different property's validated direction. We release the full protocol, all target harnesses, and the seed-replication data.

1 Introduction

Difference-of-means activation steering – adding a direction built from the mean activation difference between high- and low-property example groups to a model's residual stream – has been shown to reproduce known thermostability biology in ESM2-650M [3]. This raises a natural question for anyone who might want to use the technique for a new target property: does the strength of a scoring proxy's correlation with real experimental labels predict whether steering toward it will actually work?

We find the answer is no, and not just weakly no. Across five novel target properties tested under one pre-registered protocol, the single best-validated proxy in our entire study ($r = 0.80$ held-out, for binding affinity) produces a flat null effect at every tested intervention strength, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a clean, monotonic, artifact-checked effect that replicates across three independent random seeds. This is not an isolated anecdote: an independent, component-level result in the same model family shows the same pattern one level down. Vig et al. [4] found attention heads in protein language models that strongly align with real 3D residue contacts, explicitly flagging this as "purely associative" and untested causally. When we ablate that head and measure its actual causal effect on masked-residue prediction, the effect is statistically indistinguishable from ablating a low-correlation control head – and a full sweep of all 480 heads in the model shows the correlational top pick ranks 313th of 480 by real causal effect.

Two levels, two independent techniques (activation steering and attention-head ablation), the same lesson: a correlational signal is evidence about a model's *representations*, not about what happens when you actually *intervene*. This paper's contribution is not the steering technique itself, which we borrow unchanged from prior work, but the audit discipline needed to trust a causal claim in this domain – demonstrated concretely via two near-misses we caught in our own pipeline before they could be reported as clean results.

2 Related Work

Huang et al. [3] introduce difference-of-means activation steering for protein language models and validate it on thermostability; our work extends their harness to five new target properties under a pre-registered artifact-checking protocol, and is the first (to our knowledge) systematic test of whether proxy correlation strength predicts steering success across multiple unrelated properties.

Vig et al. [4] identify attention heads in protein language models whose attention patterns strongly align with real 3D contact maps, explicitly noting this correlation was never tested causally. We redo their finding as a genuine causal ablation test and extend it to a full 480-head sweep.

Adams et al. [1] train sparse autoencoders on ESM-2's residual stream and show, via linear probing (a correlational method), that known determinants of thermostability and subcellular localization are identifiable in SAE feature space. Their one steering experiment checks only that steering a family-specific SAE latent changes fewer residues than steering a random latent – a coherence check, not a validation against a real experimental label. Their own discussion states plainly that "investigating how to steer the model in useful ways remains an open direction for future work" and that "further research is needed to determine whether SAEs can be used as a practical tool for protein engineering." Our work answers a version of that question for a different, non-SAE steering mechanism, and specifically for the question of whether correlational validity (of either a component or a proxy) predicts causal usefulness.

3 Method: A Pre-Registered Six-Criterion Protocol

Before testing any new target property, we lock a decision rule with six criteria, applied uniformly and computed automatically at the end of each run rather than eyeballed after the fact:

1. **Beats both controls, head-to-head, with a real confidence interval.** The real steering direction must significantly beat, via direct paired bootstrap (10,000 resamples), a matched-norm random-direction control – not two separate comparisons against an unsteered baseline.

2. **Dose/scale-response, not a one-off.** The effect must show a coherent, monotonic trend across at least three steering-strength values (α), restricted to a pre-established safe operating range (see §6.1 for why this restriction matters).

3. **Survives a residue-exclusion robustness check.** If the effect is driven by generation collapsing into a narrow amino-acid composition, it is an artifact. We rerun scoring with the dominant substituted residue(s) excluded; a real effect must survive, even if diminished.

4. **Proxy pre-validated against real labels, before the steering run.** No proxy is used in a PASS decision without first being checked against real experimental labels, ideally on a held-out split.

5. **Beats the best-known existing technique**, where one exists for the target property (not applicable for a genuinely new property with no prior baseline).

6. **Adequately powered.** A minimum of 150 held-out evaluation sequences for any claim involving a new target property.

A target that fails any criterion outright is a **KILL**. A target that passes 1–2 criteria but fails on 3 or 4 is **AMBIGUOUS** – informative, but not actionable. All five novel targets in §5, plus one mechanistic follow-up on the original thermostability target (§6.1), are judged against this fixed rule.

4 Component-Level Evidence: Correlation Does Not Predict Causal Necessity

Before turning to target properties, we recap a component-level result in the same model family that motivated the property-level question. Vig et al. [4] found that a specific attention head in a protein language model shows striking correlation with real 3D residue contacts (12.9× enrichment over background on real

PDB structures), but explicitly characterized this as associative, not causal. We ablate that exact head and measure its effect on real masked-residue prediction accuracy at contact-bearing positions: the effect is statistically indistinguishable from ablating a genuinely low-correlation control head. A follow-up sweep ablating all 480 attention heads in the model, ranking each by directly measured causal effect with zero correlational information used in the ranking, confirms this was not an unlucky choice of control: Vig's correlational top pick ranks 313th of 480 by actual causal effect, and 166 of 480 heads (35%) show exactly zero measurable causal effect at all – real redundancy in the network, not a ranking artifact.

5 Target-Property-Level Evidence: Five Novel Steering Targets

We apply the protocol from §3 to five properties never previously steered in this model: binding affinity, catalytic activity, intrinsic disorder, immunogenicity, and expression yield. Each uses a real experimental dataset, a compositional scoring proxy validated against real labels before any steering run, and the identical steering/control/bootstrap harness, varying only the target dataset and proxy.

Table 1: Summary of all five novel target properties. Proxy r is Pearson correlation against real labels on a held-out split, computed before any steering run.

Target	Dataset	Proxy r	Steering result	Verdict
Binding affinity	ProteinGym DMS (RASK/DARPin)	0.80	flat null, all α	KILL
Catalytic activity	DLKcat (cross-protein)	0.22	PASS, 3/3 seeds	**PASS**
Intrinsic disorder	DisProt	0.45	PASS, 3/3 direction; 2/3 robustness	**PASS***
Immunogenicity	IEDB (4-tier)	$\leq$0.10	not run	KILL (pre-run)
Expression yield	eSol	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1, 2, 4, 6, and 2 of 3 seeds on criterion 3 (see §5.3).

Figure 1 makes this pattern visible directly: proxy validity and causal effect size show no positive relationship across the four runnable targets, and the two extremes are the opposite of what a naive "trust the best-validated proxy" heuristic would predict.

5.1 Binding Affinity: The Sharpest Null

Using a ProteinGym deep-mutational-scanning assay of KRAS binding to a DARPin binder, we build a proxy from per-position mutational-sensitivity weights learned on the training split, scoring a generated sequence by its preferential preservation of the wild-type residue at binding-critical positions. This proxy is validated far more rigorously than any other in our study: $r = 0.80$ full-set / 0.80 held-out test, confirmed with a weight-shuffle null control ($r = -0.001$), held-out position generalization ($r = 0.54$ on positions never seen during weight-fitting), and mutational-load extrapolation from single to double mutants ($r = 0.70$). Despite this, the steering effect is a flat, unambiguous null: the real-direction score is statistically indistinguishable from a matched-norm random control at every tested α, with zero degenerate sequences at any strength – ruling out generation collapse as an alternative explanation for the null.

A plausible mechanistic reason: the vector-building groups in this dataset are single-backbone point mutants (1–2 residues different from a shared 188-residue reference), giving the difference-of-means construction very little raw compositional signal to work with, independent of whether binding affinity is causally steerable in this model at all. We measure this directly – the mean per-residue amino-acid compositional distance between the low- and high-binding vector-building groups is 0.003 for this target, versus 0.02–0.05 for the three cross-protein datasets in our sweep (catalytic activity, disorder, expression yield).

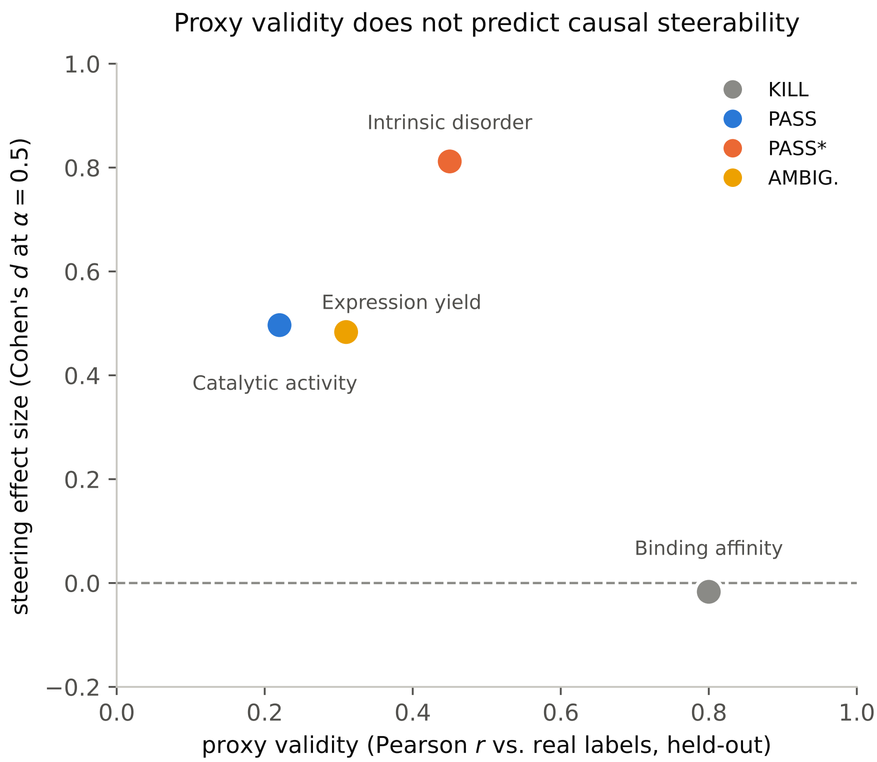

Figure 1: The four runnable targets from Table 1, plotted as proxy validity (Pearson r against real labels) against steering effect size (Cohen's d at $\alpha = 0.5$, the real-vs-random-control difference normalized by baseline standard deviation). The best-validated proxy (binding affinity, $r = 0.80$) produces the smallest effect (indistinguishable from zero); a proxy less than half as strong (catalytic activity, $r = 0.22$) produces one of the two largest effects. There is no visible positive trend.

5.2 Catalytic Activity: A Real Effect from the Weakest Proxy

Using DLKcat, a cross-protein enzyme turnover-number (kcat) dataset spanning many organisms and EC classes, we build a proxy from the fraction of glycine minus the fraction of arginine in a sequence – matching the published activity-stability tradeoff in the enzymology literature (flexible, glycine-rich active sites trade turnover for thermostability's arginine-mediated ion pairs). This proxy validates at $r = 0.22$ held-out – the weakest of the four runnable targets in our sweep – yet produces a clean, monotonic dose-response effect, significant at every tested α in the safe range, that survives residue-exclusion (effect retains roughly one third of its magnitude after excluding the two dominant substituted residues, one of which is literally one of the proxy's own two defining terms). We reran the entire pipeline with two additional random seeds for both the train/vector-building split and the random-control direction draw: all six criteria hold on all three seeds, with the same order of magnitude and monotonic shape each time. This is the one target in our sweep with fully unanimous seed-robustness.

5.3 Intrinsic Disorder: A Real Effect With a Seed-Sensitive Artifact Check

Using DisProt (real per-residue intrinsic-disorder annotations, reduced to a per-sequence scalar), we build a proxy from the published TOP-IDP amino-acid disorder-propensity scale [2], which validates at $r = 0.45$ held-out – the strongest of any target in our sweep – confirmed additionally at the per-residue level (AUC 0.71 against real per-residue disorder labels via a sliding window). The steering effect is significant and monotonically dose-responsive on every one of three independent seeds. However, the residue-exclusion robustness check specifically is seed-sensitive: it passes on two of three seeds (36–42% of effect magnitude retained, confidence interval excludes zero) and fails on the third (10% retained, confidence interval crosses zero). The effect's existence and direction are seed-stable; the artifact check that rules out pure compositional collapse is not uniformly so. We report this as a majority pass with an explicit caveat, not a unanimous clean result – and as evidence that different criteria within the same protocol can carry different seed-sensitivity, which no single run can reveal.

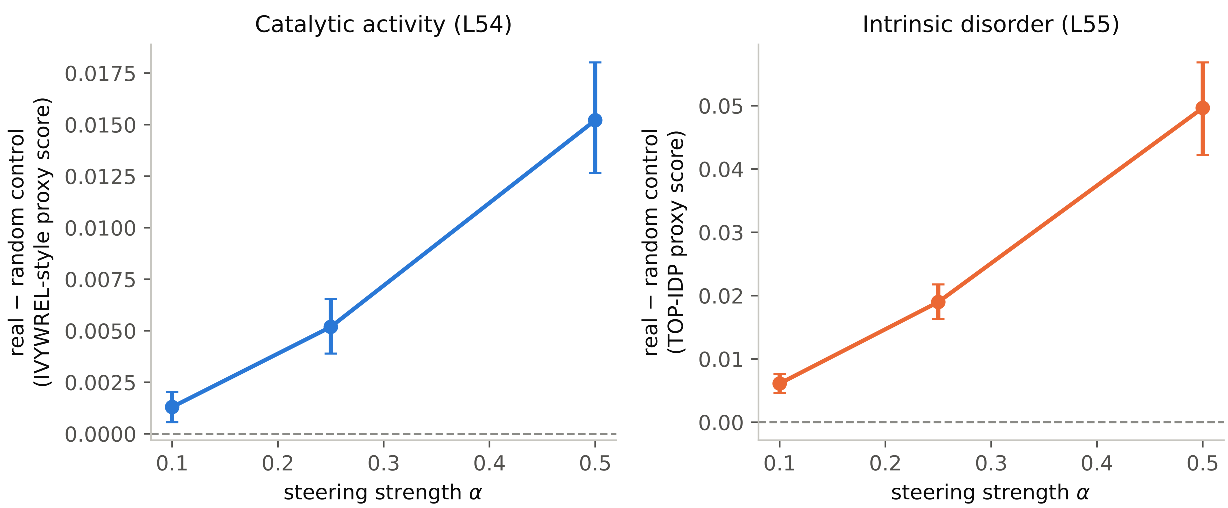

Figure 2: Dose-response for the two targets that pass criterion 2: the real steering direction's score minus the matched-norm random control's score, at each tested α in the pre-established safe operating range. Both show a clean, monotonic increase with error bars (95% bootstrap CIs) that stay bounded away from zero at every point.

5.4 Immunogenicity: Killed at the Proxy Gate, With a Named Confound

For immunogenicity, we evaluate candidate compositional proxies across four tiers of increasing fidelity to the real target: peptide MHC-II binding affinity (a surrogate), mass-spectrometry-confirmed peptide presentation, real T-cell response rate (the actual endpoint), and full-length source antigens (the regime our steering pipeline actually operates in). Proxies that look genuinely predictive on the surrogate binding-affinity tier ($r = 0.37$–0.43) collapse to near-chance on the real T-cell-response endpoint ($r \leq 0.10$) and, on full-length antigens, one proxy that appears to reach $r = 0.38$ turns out to be a source-organism confound: organism identity alone explains 58% of the apparent signal's variance, and the same proxy's correlation flips to -0.32 once organism identity is held out of cross-validation. No proxy survives to criterion 4, so no steering script was written and no model run was performed – an intended KILL under the protocol, not missing work.

5.5 Expression Yield: An Ambiguous Result Explained by Vector Geometry

Using eSol (real E. coli soluble-expression-yield measurements, verified statistically distinct from an unrelated, previously-tested solubility dataset via their 441 overlapping sequences showing zero correlation between the two labels), we build a proxy from the absolute deviation of net charge from neutrality, following a published recombinant-solubility discriminant [5]. This proxy validates at $r = 0.31$–0.34 held-out and produces a significant, dose-responsive steering effect – but the effect collapses to statistical insignificance under residue-exclusion (one of the two dominant substituted residues is literally one of the proxy's own defining terms), the same compositional-collapse artifact shape our protocol is designed to catch.

Rather than discarding this as unexplained noise, we compute the cosine similarity between this target's steering vector and every other target's, layer by layer. We find substantial, non-trivial overlap with the intrinsic-disorder target's vector (§5.3): $+0.38$ on the full 33-layer concatenation, rising to $+0.56$ to $+0.67$ at the three deepest layers. This is not coincidental – disordered protein regions are independently documented in the literature to reduce solubility and expression yield – and reframes the result: rather than independent evidence of a sixth steerable property, this target's steering vector is best understood as a partial, noisier echo of disorder's already-validated real effect, filtered through a proxy that happens to be more prone to compositional collapse than TOP-IDP.

6 Self-Correction Case Studies

Two results in this study were nearly reported incorrectly before additional scrutiny caught the error. We describe both explicitly rather than silently absorbing the fix, since the same failure modes are likely to recur in similar pipelines.

6.1 A Spurious PASS From Comparing Arms at an Unsafe Alpha

As a mechanistic follow-up to the original thermostability reproduction, we tested whether restricting the steering intervention to only the five layers previously identified as causally necessary (via both a sufficiency and an independent leave-one-out necessity sweep) preserves the effect of steering all 33 layers. A first implementation selected its best-performing α for the head-to-head comparison from the full tested range, including $\alpha \geq 1.0$ – and reported a clean PASS: the 5-layer subset appeared to outperform the full 33-layer intervention at $\alpha = 2.0$. This was an artifact of comparison, not a real advantage: at that α, the 33-layer intervention had already fully collapsed into degenerate, repetitive output (zero non-degenerate evaluation sequences remained), while the 5-layer subset had not yet collapsed. The subset "won" only because it broke down later, not because it steered harder in any comparable regime.

We fixed this by restricting the α range used for this comparison (and for criterion 2's dose-response check more generally) to a pre-established safe operating window in which neither arm has collapsed, and reran the full experiment end to end rather than patching the already-computed numbers. The corrected result: the 5-layer subset does produce a real, significant, dose-responsive, residue-robust effect, but retains only 43–59% of the full 33-layer effect size at every α where both arms are trustworthy – a real, informative partial result, not the unqualified success the uncorrected analysis suggested.

6.2 A Criterion That Only Passes on Two of Three Seeds

As described in §5.3, our intrinsic-disorder target's residue-exclusion robustness check (criterion 3) passes on two of three independently-seeded reruns and fails on the third, even though the underlying effect's existence and direction (criteria 1–2) are stable across all three. Every result in this study before this check used a single seed for the train/vector-building split and the random-control direction draw; a single unlucky (or lucky) seed could have reported either a false unanimous PASS or a false KILL on this specific criterion. We recommend that any activation-steering pipeline reporting a criterion-specific robustness check rerun with at least two additional seeds before treating that specific criterion's verdict as settled, even if the headline effect (existence and direction) is stable.

7 Discussion

What "causal" means here. Every steering result in this paper is causal about the model's activation space: a real intervention on the forward pass, evaluated against a matched-norm random-direction control and a residue-exclusion artifact check, on generated (not merely scored) sequences. None of these results are validated against a real biological assay confirming that a generated, steered sequence actually possesses the intended property at the wet-lab level. This distinction should be made explicit in any application of this technique: a PASS here means "causally steers a validated compositional proxy," not "causally engineers the underlying biological property."

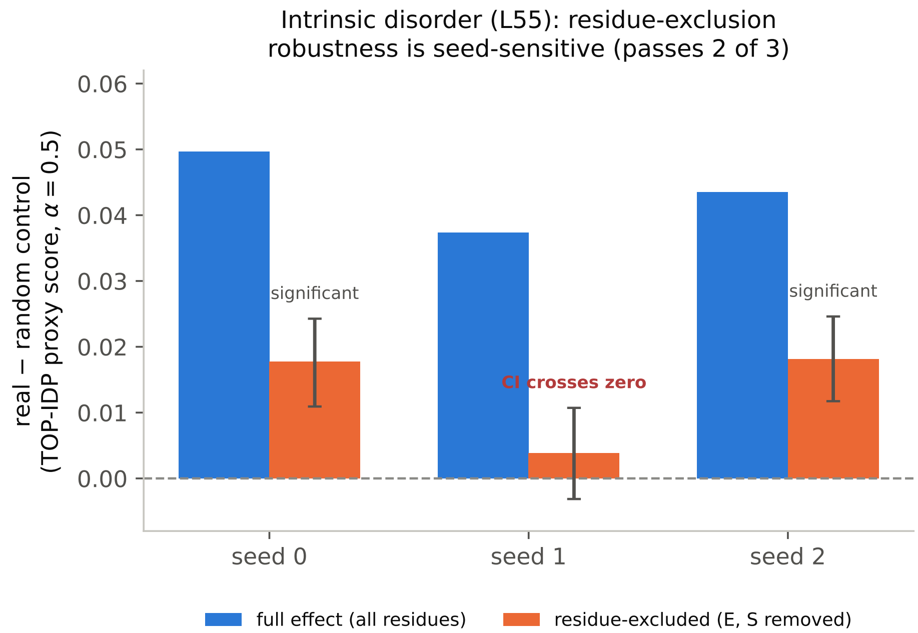

Figure 3: The intrinsic-disorder target's effect (blue) and the same effect after the two dominant substituted residues are excluded (orange, with 95% bootstrap CIs), across three independent seeds. The full effect is stable across all three; the excluded-residue effect is significant on seeds 0 and 2 but its confidence interval crosses zero on seed 1 – the same underlying effect, the same criterion, three different verdicts on that one criterion.

The vector-geometry diagnostic generalizes. The check that explained our expression-yield result (§5.5) – computing cosine similarity between a new target's steering vector and every previously-validated target's vector – is cheap (one additional embedding pass, not a full steering sweep) and, we suggest, worth running by default whenever more than one target property is tested in the same study. An AMBIGUOUS result that turns out to be geometrically explained by a different, already-validated effect is a qualitatively different finding than an AMBIGUOUS result that is not.

Limitations. All experiments use a single model (ESM2-650M) and a single model size; we did not test whether these findings hold at other scales. All proxies are compositional heuristics, not learned predictors, chosen for interpretability and speed rather than maximal predictive power – a stronger, non-compositional proxy might behave differently. Each target's final proxy formula was selected as the best of several candidates (typically 7-9) scored on the same held-out split whose r is then reported as the validation number, rather than via a separate train/select/test split – a garden-of-forking-paths bias that optimistically inflates every reported proxy-r in the same direction. This does not by itself explain the central finding (binding affinity's proxy remains the best-validated and its steering effect remains null under any plausible correction), but the absolute r values should be read as upper bounds. We also ran a large number of significance tests across the study (5 targets $\times$ 5 alphas, with 3-seed reruns for two targets, plus L48/L49's 480-head sweep) without a multiple-comparisons correction; this is partially self-mitigated by requiring the same criterion to hold across multiple alphas and seeds simultaneously for a PASS, but individual borderline criteria (e.g. L55's residue-exclusion check, significant on 2 of 3 seeds) sit close enough to the noise floor that this is a real caveat, not a formality. We tested seed-robustness for two targets (catalytic activity, intrinsic disorder) but not all five; the seed-sensitivity finding in §5.3 may or may not generalize to the other three. We did not run any structural or functional plausibility check (e.g. predicted-fold confidence) on generated sequences, so a PASS result does not rule out that steered sequences, while scoring well on the proxy, are otherwise implausible proteins.

8 Conclusion

Correlational validity – of an attention head's alignment with a real biological signal, or of a scoring proxy's correlation with real experimental labels – does not predict causal effect when actually tested, at two different levels of the same class of models. The practical implication for anyone applying activation steering to a new protein property is direct: proxy validation strength is not a reliable filter for which targets are worth attempting, and a single steering run, even one that clears every criterion, should not be trusted without at least a seed-robustness check on results whose downstream use matters.

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